

Complexities of Communication, Cognition, and Similarities to NLP

Introduction

Denis Villeneuve's *Arrival* (2016) opens as a story about alien arrival and first contact, but evolves quickly into a study on communication, cognition, and what it means to truly grasp language. Protagonist Dr. Louise Banks is a linguist who is tasked with deciphering the language of the "heptapods," a species whose sudden appearance causes both apprehension and fear. The character's challenge is not just to translate with the extraterrestrials are saying but to comprehend how they think. This is akin to humans in the real world who attempt to build intelligent systems that naturally understand human language.

Arrival acts as an analogy for the field of Natural Language Processing (NLP), the branch of AI focused on enabling machines to interpret and generate human language. Like the character Dr. Banks, NLP models begin their work amid uncertainty, often lacking the data, context, or structure to make sense of the language input. Further, the film dramatizes three core challenges in modern NLP: the difficulty of learning from limited data, the gray area of meaning shaped by cultural framing, and the limits of linear language processing. Through its resolution, *Arrival* demonstrates how language shapes not just communication but perception — a truth that holds significance for both the human brain and artificial intelligence systems learning to "see" the world through words.

Discussion

Dr. Banks begins with no shared dataset or frame of reference. Her process — pointing to herself and saying "Louise," then writing "Human" on a whiteboard — represents the slow, high-stakes work of building meaning from scratch. This is the linguistic equivalent of starting an NLP model in a low-resource environment, where there are few examples from which to learn. Like zero-shot or few-shot learning, progress in *Arrival* depends on iteration, observation, and trust. Both the linguist and the machine face the same constraint: true understanding requires immense computational effort and time when there is no prior data to guide interpretation.

The film's central conflict emerges when the heptapods' phrase "offer weapon" is misinterpreted as "use weapon." The misunderstanding reflects how meaning depends not only on words but on worldview. While Dr. Banks interprets the message as an invitation to collaboration, the Chinese military sees it as a threat. This divergence underscores how cultural framing shapes linguistic interpretation. In NLP, this mirrors Word Sense Disambiguation — the task of

determining which meaning of a word is intended based on context. Just as Dr. Banks must understand the heptapods' cooperative intent, AI models must learn to differentiate between meanings like "bank of a river" and "financial bank." Without sufficient cultural or contextual data, even advanced systems can misinterpret signals in ways that lead to serious consequences.

The heptapods' written language, composed of circular logograms, defies human conventions. Each symbol communicates an entire thought simultaneously, rather than word by word. This form of expression mirrors the evolution of NLP itself — from Recurrent Neural Networks, which processed text sequentially, to Transformer models such as BERT and GPT, which use attention mechanisms to analyze all words in context simultaneously. The heptapod system represents the ultimate vision of such models: non-linear comprehension that captures the full semantic weight of an idea all at once, rather than parsing it in parts.

Modes of Communication in Arrival: Strengths and Limitations

Verbal speech initially proves ineffective between humans and heptapods. While spoken language allows for emotion and immediacy among humans, it fails here because the biological structures of sound production are incompatible. Verbal communication, while powerful for human connection, is fragile — easily misheard, forgotten, and dependent on environmental factors like proximity and sound clarity.

Dr. Banks' whiteboard breakthrough marks the transition to a shared symbolic system. By writing "Human" and associating it with visual context, she establishes a supervised form of learning — a process analogous to dataset labeling in machine learning. Written symbols provide permanence and allow both sides to reference meaning repeatedly. However, the method is slow and susceptible to misunderstanding due to missing context. Its linear format cannot capture the complexity of the heptapods' non-linear concepts, making it both a triumph of accessibility and a limitation of form.

China's use of mahjong introduces a rigid, rule-based framework that efficiently conveys concepts of strategy, turn-taking, and victory. However, it also imposes a zero-sum framework, shaping every exchange as competitive rather than cooperative. The result is bias — a misalignment between method and meaning. Similarly, in NLP, when models are trained exclusively on data shaped by one cultural or philosophical lens, they inherit that bias, leading to limited interpretive flexibility.

Dr. Banks' decision to remove her hazmat suit and place her hand on the glass barrier represents a powerful act of nonverbal communication. It conveys trust and vulnerability — a physical form of connection that transcends symbols. While this act fosters empathy, it carries risk and lacks precision. Like multimodal AI systems that integrate vision and text, this moment expands the communication channel but cannot transmit abstract meaning without shared understanding.

Face-to-face exchanges allowed for emotion, tone, and correction in real time, making them the richest form of human communication. However, they are costly, limited in scale, and dependent on access and safety. The parallel to NLP is clear: models that rely on human fine-tuning for every task cannot scale effectively. Trust, efficiency, and shared context must be built into the system itself.

Reflection: Rewiring the Brain and Retraining the Model

The climax of *Arrival* reveals that learning the heptapod language transforms not just communication but cognition. Dr. Banks begins to perceive time non-linearly, understanding past and future as simultaneous. This embodies the Sapir-Whorf hypothesis, which proposes that language shapes thought. Her brain rewires to accommodate new dimensions of understanding — a human metaphor for neuroplasticity.

In NLP, the analogy is direct. Exposing models to new linguistic structures, cultural perspectives, and multimodal data expands their contextual depth. Just as Louise's mind adapts to perceive new forms of meaning, AI systems trained across diverse datasets become more flexible and nuanced. This variety does not simply add information; it transforms the model's architecture, enabling broader, more holistic comprehension.

Conclusion

Arrival reframes communication as both a linguistic and cognitive act. The film's layered depiction of language — as sound, symbol, gesture, and perception — shares similar layered challenges of NLP, where meaning emerges only through context and diversity of input. Dr. Banks' journey from decoding words to reshaping thought illustrates the ultimate goal of artificial intelligence: to move beyond syntax toward understanding.

The lesson extends beyond the film. Language does not merely describe reality; it constructs it. For humans, learning a new language enhances neuroplasticity — the ability to see and think differently. For NLP models, exposure to varied linguistic and cultural data performs the same function, rewiring artificial architectures toward greater adaptability. Both humans and machines evolve through communication. *Arrival* shows us that the key to intelligence — biological or artificial — is not just learning to speak, but learning to see the world anew.

Citations

Buono, R. A., M. Nygren, and N. Bianchi-Berthouze. "Touch, Communication and Affect: A Systematic Review on the Use of Touch in Healthcare Professions." *Systematic Reviews*, vol. 14, no. 42, 2025, <https://doi.org/10.1186/s13643-025-02769-4>. Accessed 11 Oct. 2025.

Ribeiro, Tatiana. "Letterforms as Visual Communication Elements." CUNY Academic Works, New York City College of Technology, 4 Dec. 2019, https://academicworks.cuny.edu/ny_pubs/517/

. Accessed 11 Oct. 2025.

“Science of Noise.” Tumblr, post titled “A Desire for More Cows,”
<https://science-of-noise.tumblr.com/post/154009033405/a-desire-for-more-cows>
. Accessed 11 Oct. 2025.

Stanford University. “The Power of Language: How Words Shape People, Culture.” Stanford News, 22 Aug. 2019,
<https://news.stanford.edu/stories/2019/08/the-power-of-language-how-words-shape-people-culture>. Accessed 11 Oct. 2025.

Villeneuve, Denis, director. Arrival. Paramount Pictures, 2016. Apple TV+, Accessed 11 Oct. 2025.